Project Proposal: Visualizing the Global Energy Transition
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Abstract

This proposal describes a visualization project exploring the global energy transition using two open datasets from Our World in Data: a comprehensive energy dataset (23,232 rows, 130 attributes) and a CO_2 emissions dataset (50,411 rows, 79 attributes). The merged dataset covers 220 countries from 1900 to 2025 with 201 attributes spanning energy consumption by source, electricity generation mix, renewable deployment, greenhouse gas emissions, and socioeconomic indicators. Four interactive visualizations are proposed: (1) an animated Sankey diagram of national energy flows, (2) small-multiple stacked area charts of regional electricity mix, (3) a Gapminder-style animated scatterplot relating wealth to decarbonization, and (4) interactive parallel coordinates for multi-dimensional country energy profiling. The project targets clients in various sectors like government ministries, development banks, automotive/EV maufacturers and climate investors who need to assess, compare, communicate and interpret national and regional progress through the energy transition.

1 Introduction

The global energy transitional shift from fossil fuels to low-carbon energy sources is one of the defining challenges of the 21st century. Tracking this transition across countries and over time requires integrating data on energy supply, electricity generation, emissions, and economic development. Effective visualization of these interrelated dimensions can reveal patterns, outliers, and trajectories that tabular data alone cannot communicate.

1.1 Datasets

This project uses two openly available datasets maintained by Our World in Data (OWID), both released under Creative Commons (CC) licenses:

1. **OWID Energy Dataset** (`owid-energy-data.csv`): 23,232 rows × 130 columns. Covers 220 countries and 94 aggregate regions from 1900–2025. Contains annual data on primary energy consumption by source (coal, oil, gas, nuclear, hydro, solar, wind, biofuel, other renewables), electricity generation and mix, per-capita and share-of-total metrics, and socioeconomic indicators (GDP, population).

   Source: Energy Institute Statistical Review, Ember Yearly Electricity Data, and U.S. EIA.

2. **OWID CO_2 and Greenhouse Gas Emissions Dataset** (`owid-co2-data.csv`): 50,411 rows × 79 columns. Covers 254 countries from 1750–2024. Contains CO_2 emissions by fuel type (coal, oil, gas, cement, flaring), consumption-based vs. production-based emissions, cumulative emissions, methane, nitrous oxide, total GHG, and temperature-change contributions.

   Source: Global Carbon Budget and Jones et al.

The two datasets share `country` and `year` as merge keys. A left join of the energy dataset with the 71 CO_2-only columns yields a merged dataset of 23,232 rows × 201 columns, with 17,991 rows having matched records in both sources.

1.2 Client

The client may be a strategic analyst in a national energy planning ministry (e.g., Natural Resources Canada). The analyst is responsible for advising senior officials on domestic energy infrastructure investment and national renewable energy target-setting. Specifically, the analyst needs to: (a) understand how the country's energy mix has evolved structurally over the past three decades and where it stands today, (b) benchmark the country's transition progress against peer nations at similar stages of economic development, identifying leaders to emulate and patterns to avoid, (c) compare regional transition trajectories worldwide to assess where global momentum is building and where it is stalling, and (d) profile countries across multiple energy and emissions dimensions simultaneously to identify peer clusters and inform bilateral cooperation strategies. The visualizations are designed to directly support these planning and benchmarking tasks, enabling the analyst to move from high-level structural overview to targeted country-level comparison.

2 Preliminary Data Analysis (*What*)

Following Munzner's *what* methodology, we characterize the datasets in terms of data types, dataset types, and attribute types.

2.1 Dataset Types

- **Tables**: Both datasets are flat tables (items × attributes). The energy dataset has 23,232 items (country-year observations) and 130 attributes. The CO_2 dataset has 50,411 items and 79 attributes. The merged table has 201 attributes.
- **Temporal**: Both datasets are longitudinal, with `year` serving as an ordered key attribute. The energy data spans 1900–2025; CO_2 data spans 1750–2024. The most analytically useful range is 1990–2023, where data completeness exceeds 90% for electricity-related attributes.
- **Spatial/Network (implicit)**: Countries have geographic positions (not encoded directly in the data, but available via ISO codes). The aggregate regions (Africa, Asia, Europe, North America, South America, Oceania) provide a hierarchical geographic structure.

2.2 Attribute Types

Table 1 summarizes the key attributes used in the project visualizations.

2.3 Data Sufficiency

The merged dataset provides ample scale for visualization: 23,232 rows, 201 attributes, 220 countries, and temporal coverage exceeding a century. For the primary analysis period (2000–2023), data completeness for electricity generation columns exceeds 92%, and for energy consumption columns exceeds 95%. The CO_2 merge yields matched data for 17,991 country-year pairs. The diversity of attribute types—categorical (country, continent), ordered sequential (year), and quantitative (consumption, emissions, GDP, shares)—supports a wide range of visual encodings.

3 Client Questions (*Why*)

The following three questions frame the project goals, expressed in terms that reflect the client's analytical needs. Each question suggests different ways of examining the data and supports distinct visualization designs.

3.1 Question 1: How has the structure of energy supply changed over time, and how do these shifts differ between countries?

This question targets the **temporal evolution of energy mix composition** at both global and national levels. It is deliberately broad, asking about structural change rather than a specific metric, which leaves

Table 1: Key attributes and their types (Munzner classification)

Attribute	Type	Order	Source
`country`	Categorical	Unordered	Energy
`year`	Ordered (Key)	Sequential	Both
`iso_code`	Categorical	Unordered	Energy
`population`	Quantitative	Sequential	Energy
`gdp`	Quantitative	Sequential	Energy
`coal_consumption`	Quantitative	Sequential	Energy
`oil_consumption`	Quantitative	Sequential	Energy
`gas_consumption`	Quantitative	Sequential	Energy
`nuclear_consumption`	Quantitative	Sequential	Energy
`hydro_consumption`	Quantitative	Sequential	Energy
`solar_consumption`	Quantitative	Sequential	Energy
`wind_consumption`	Quantitative	Sequential	Energy
`renewables_share_energy`	Quantitative	Sequential	Energy
`fossil_share_energy`	Quantitative	Sequential	Energy
`carbon_intensity_elec`	Quantitative	Sequential	Energy
`energy_per_capita`	Quantitative (Derived)	Sequential	Energy
`*_electricity` (9 sources)	Quantitative	Sequential	Energy
`co2`	Quantitative	Sequential	CO_2
`co2_per_capita`	Quantitative	Sequential	CO_2
`renewables_share_elec`	Quantitative	Sequential	Energy
`gdp_per_capita` (derived)	Quantitative	Sequential	Derived
continent (derived)	Categorical	Unordered	Derived

design room to choose between absolute quantities and proportional shares, and between single-country deep dives and multi-country comparisons.

Munzner *why*: ACTION: Discover trends, compare patterns across countries. TARGET: Trends in the distribution of energy sources over time.

This question motivates Visualizations 1 (Sankey) and 2 (stacked area), which address structural composition from complementary angles.

3.2 Question 2: Is there a relationship between national wealth and the adoption of renewable energy, and which countries outperform or underperform their economic bracket?

This question probes a **specific bivariate relationship**—GDP per capita vs. renewable energy share—while also asking about outliers. It is more targeted than Question 1, directing attention to a particular pair of attributes and to the identification of interesting individual cases.

Munzner *why*: ACTION: Identify correlation, locate outliers. TARGET: Correlation between wealth (GDP per capita) and decarbonization (renewable share), with CO_2 as a third quantitative dimension.

This question directly motivates Visualization 3 (animated scatterplot).

3.3 Question 3: What multi-dimensional energy profiles characterize different groups of countries, and can we identify clusters or archetypes?

This question asks about **high-dimensional country characterization** across multiple energy and emissions attributes simultaneously. It is exploratory—the analyst does not know in advance what clusters exist—but it is constrained to a specific recent year and a defined set of dimensions.

Munzner *why*: Action: Explore, identify clusters, find outliers. Target: Multi-attribute profiles across countries, seeking patterns across renewable share, fossil share, carbon intensity, energy per capita, CO_2 per capita, and GDP per capita.

This question motivates Visualization 4 (parallel coordinates).

4 Proposed Visualizations (*How*)

Four visualizations are proposed, each addressing one or more client questions. All are implemented in Plotly (Python) and exported as interactive HTML files.

4.1 Visualization 1: Animated Sankey — Energy Source Flows

Goal: Show how a country's primary energy consumption is distributed across sources and categories (fossil, nuclear, renewable), animated over time.

Design: A Sankey diagram with source nodes on the left (coal, oil, gas, nuclear, hydro, solar, wind, biofuel, other renewables) and category nodes on the right (Fossil Fuels, Nuclear, Renewables). Link widths encode consumption in TWh. A year slider and play/pause buttons animate the diagram from 1990 to 2023, revealing structural shifts.

Relation to questions: Directly addresses Question 1 (structural change over time). The Sankey makes proportional shifts viscerally visible—shrinking fossil flows and growing renewable flows are immediately apparent.

Marks and channels: Links (marks) encode flow magnitude via width (channel). Nodes use color to distinguish sources and categories. Animation encodes time.

Creativity: Animated Sankey diagrams are uncommon in standard visualization galleries. The combination of flow visualization with temporal animation goes beyond standard bar/line/scatter idioms.

***How* elements**: Encode (link width → quantity, color → category), Manipulate (animate via slider, select country), Facet (source-to-category flow structure), Reduce (aggregate by energy category).

4.2 Visualization 2: Small-Multiple Stacked Area — Regional Electricity Mix

Goal: Compare how the electricity generation mix has evolved across six world regions (Africa, Asia, Europe, North America, South America, Oceania).

Design: A 2×3 grid of stacked area charts, one per region. Each panel shows electricity generation (TWh) from 1990–2023, stacked by source: coal, gas, oil, nuclear, hydro, wind, solar, biofuel, other renewables. Shared color legend across panels; region-specific y-axes to preserve local detail.

Relation to questions: Addresses Question 1 (how has the mix changed?) with an emphasis on cross-regional comparison. The small-multiple layout enables the analyst to scan all six regions in a single view.

Marks and channels: Area marks encode generation quantity. Vertical position (stacking) and color both encode source type. Horizontal position encodes time. Spatial faceting encodes region.

***How* elements**: Encode (area → quantity, color → source, position → time), Facet (small multiples by region), Reduce (pre-aggregated regional totals from OWID).

4.3 Visualization 3: Animated Scatterplot — Wealth vs. Decarbonization

Goal: Reveal the relationship between economic development and renewable energy adoption, and identify countries that outperform or underperform.

Design: A Gapminder-style bubble chart. X-axis: GDP per capita (log scale). Y-axis: renewable share of primary energy (%). Bubble size: total CO_2 emissions. Color: continent. Animated by year (2000–2023) with play/pause controls. Hover shows country name and exact values.

Relation to questions: Directly addresses Question 2 (wealth vs. decarbonization, outlier identification). The animation adds Question 1's temporal dimension—the analyst can watch countries migrate across the wealth-decarbonization space over two decades.

Marks and channels: Point marks (bubbles). Position encodes GDP and renewable share. Size encodes CO_2. Color encodes continent. Animation encodes time.

Creativity: The Gapminder idiom is well-known but rarely applied to energy transition data. Using CO_2 as the size dimension creates a triple encoding that reveals the "big emitters" immediately. The merged dataset (energy + CO_2) enables this multi-source design.

How **elements**: Encode (position → GDP and renewable share, size → CO_2, color → continent), Manipulate (animate, hover, filter), Reduce (filter to countries with complete data, clip outlier bubble sizes).

4.4 Visualization 4: Interactive Parallel Coordinates — Country Energy Profiles

Goal: Enable exploratory analysis of multi-dimensional country energy profiles for a single year, revealing clusters and outliers across six dimensions simultaneously.

Design: A parallel coordinates plot for year 2022. Six axes: renewable share of electricity (%), fossil share of energy (%), carbon intensity of electricity (gCO_2/kWh), energy per capita (kWh), CO_2 per capita (tonnes), GDP per capita (USD). Each line is a country, colored by continent. Interactive brushing on any axis filters the display.

Relation to questions: Directly addresses Question 3 (multi-dimensional profiling, cluster identification). Brushing enables the analyst to isolate, for example, high-GDP countries with low carbon intensity, or African nations with high renewable share but low energy access.

Marks and channels: Line marks connect axis intersections for each country. Position on each axis encodes the respective attribute value. Color encodes continent. Interactive brushing serves as a filter.

How **elements**: Encode (axis position → attribute value, color → continent), Manipulate (brush/filter on any axis), Reduce (single year snapshot, percentile-based axis ranges to handle outliers).

Attributions

Datasets:

Hannah Ritchie, Pablo Rosado, Edouard Mathieu, and Max Roser (2023). "Energy" and "CO_2 and Greenhouse Gas Emissions." Published online at `OurWorldInData.org`.

Retrieved from `https://github.com/owid/energy-data` and `https://github.com/owid/co2-data`. Licensed under CC BY 4.0.

The energy dataset incorporates data from the Energy Institute Statistical Review of World Energy, Ember Yearly Electricity Data, and the U.S. Energy Information Administration. The CO_2 dataset incorporates data from the Global Carbon Budget (Friedlingstein et al.) and Jones et al.

Tools: Plotly (Python), Pandas, NumPy. Visualizations rendered as interactive HTML.